

AKSHAT GUPTA

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SUMMARY

Software Developer with hands-on experience designing AI-powered applications, scalable backend systems, and full-stack software using Java, Python, Spring Boot, FastAPI, React, MongoDB, and AWS. Built production-ready projects including an AI Code Optimizer, Adaptive RAG system, and hotel booking backend, with expertise in REST APIs, Retrieval-Augmented Generation (RAG), authentication, and cloud technologies.

EDUCATION

Bachelor of Science in Information Technology

2023 - 2026

Thakur College of Science & Commerce, Mumbai University

TECHNICAL SKILLS

Languages	Java, Python, JavaScript, SQL
Frontend	React.js, HTML5, CSS3, Bootstrap, Vite
Backend	Java EE, Servlets, JSP, JDBC, Spring Boot, FastAPI, REST APIs, JWT Authentication, OAuth 2.0, WebSockets
Artificial Intelligence	Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), LangGraph, Prompt Engineering, Generative AI, AI Agents
Databases	MongoDB, MySQL, FAISS
Cloud & DevOps	AWS, Git, GitHub, CI/CD (GitHub Actions)

EXPERIENCE

Software Development Intern

Jan 2026 – Jun 2026

SDAC Infotech

Mumbai, India

- Developed a Hybrid ERP application using Java EE, Servlets, JSP, JDBC, HTML, CSS, and Bootstrap, delivering production-ready application modules.
- Designed and queried MySQL database schemas, integrated JDBC connectivity, and configured an Apache server for application deployment.
- Integrated Generative AI tools into the development workflow and deployed the completed application to a live environment on AWS.
- Leveraged Git and GitHub for version control, collaborative development, and code management throughout the internship.

SELECTED PROJECTS

Adaptive Retrieval-Augmented Generation (Adaptive RAG)

[GitHub](#)

- Built an adaptive RAG system that dynamically routed user queries between document retrieval, web search, and direct LLM responses.
- Designed a LangGraph workflow featuring query routing, document relevance grading, response verification, and self-correcting regeneration loops to improve factual accuracy.
- Developed a FastAPI backend integrated with MongoDB, FAISS vector search, JWT authentication, and 29 automated tests executed through CI pipelines.
- Technologies: LangGraph, FastAPI, MongoDB, FAISS, JWT, Python, CI/CD.

AI Code Optimizer

[GitHub](#)

- Developed a full-stack AI-powered code optimization platform integrating OpenAI, Claude, and Gemini to analyze, optimize, and compare source code.
- Engineered a secure sandbox-based verification engine that executed original and AI-generated code, validated functional equivalence, and benchmarked execution time and memory usage.
- Implemented automated secret detection using 20+ security patterns with SARIF report generation for identifying exposed API keys, tokens, and insecure coding practices.
- Architected a modular FastAPI backend with 16 API modules supporting authentication, analytics, GitHub integration, OAuth login, team collaboration, and WebSocket-based live synchronization.
- Technologies: React, FastAPI, MongoDB, JWT, OAuth 2.0, WebSockets, OpenAI API, Claude API, Gemini API.

Airbnb Backend System

GitHub

- Developed a scalable hotel booking backend using Java and Spring Boot following Controller-Service-Repository architecture.
- Implemented pessimistic database locking to prevent concurrent room overbooking during high-demand reservation scenarios.
- Applied the Strategy Design Pattern for dynamic pricing and integrated Stripe payment APIs with webhook-based payment confirmation and refund handling.
- Developed RESTful APIs for hotel management, booking workflows, payment processing, and inventory management.
- Technologies: Java, Spring Boot, MySQL, Stripe API, REST APIs, JPA/Hibernate.

CERTIFICATIONS

• Oracle Cloud Infrastructure 2025 AI Foundations Associate – Oracle Credential ID: 102987448OCI25AICFA	Oct 2025
• McKinsey.org Forward Program – McKinsey.org	Dec 2025
• Generative AI Mastery Workshop – NxtWave & OpenAI Academy	Aug 2025
• Google AI Essentials Specialization – Google & Coursera Verify credential	Jun 2026
• Aspire Leaders Program 2026 – Aspire Institute	Jul 2026
• Software Development Internship – SDAC Infotech	Jun 2026

LEADERSHIP & ACTIVITIES

- Completed the **Aspire Leaders Program 2026**, strengthening leadership, communication, collaboration, and critical thinking through a 40-hour global learning experience.
- Actively develop AI-powered applications, backend systems, and full-stack software while maintaining technical projects on GitHub.